Explainable Ensemble Models for Spatio-Temporal Dengue Outbreak Forecasting in Bangladesh

***Abstract*—Dengue fever is a significant public health issue in Bangladesh, characterized by strong seasonality, spatial variability across the eight administrative divisions and non-linear associations between climatic drivers and case incidence. Classical regression and compartmental models are often inadequate for such high-dimensional, zero-inflated surveillance data, limiting their usefulness for early warning. This paper proposes an explainable ensemble-learning framework for daily, division-wise dengue forecasting that integrates dengue surveillance from the Directorate General of Health Services (DGHS) with high-resolution meteorological time series from Open-Meteo and ERA5 for January 2022–October 2025. A leakage-aware preprocessing pipeline handles missing values, zero inflation and temporal misalignment and is complemented by advanced feature engineering using autoregressive lags, rolling-window statistics, Fourier-based seasonal encodings and climate indices such as Heat Index and Rain–Temperature Index. Six regression models, including Ridge Regression, Random Forest, Support Vector Regression and gradient-boosting ensembles (LightGBM, CatBoost and XGBoost), are benchmarked under a chronological train–test split and evaluated using MAE, RMSE, RMSLE and R^2. After hyperparameter optimization, LightGBM achieves the best performance with $R^2 = 0.900$, MAE = 8.927 and RMSE = 20.094 on the 2025 test period, outperforming linear and non-ensemble baselines. Shapley Additive explanations (SHAP) reveal that recent dengue incidence and short-term rolling averages are the dominant predictors, followed by seasonal and climatic variables, providing epidemiologically consistent insight into dengue transmission drivers. The framework offers a transparent, data-driven decision-support tool for dengue early warning and resource planning in Bangladesh.**

***Index Terms*—Dengue, Explainable AI, Ensemble Learning, Time Series Forecasting, Bangladesh.**

I. Introduction

Dengue fever represents a major public health challenge in tropical and subtropical regions, including Bangladesh, where climatic and demographic conditions favour repeated outbreaks [1], [2]. In recent years, Bangladesh has experienced record-breaking dengue seasons, with large surges centered in Dhaka but increasingly affecting all administrative divisions [2]. This expansion underscores the need for predictive methods that capture both temporal dynamics and regional variation in risk.

Transmission dynamics are driven by *Aedes* mosquitoes, whose breeding behaviour and vectorial capacity depend on climatic conditions such as temperature, rainfall and humidity [3]. Monsoon rainfall, escalating temperatures and rapid, often unplanned urbanization jointly create favourable conditions for mosquito proliferation and lead to highly seasonal, spatially heterogeneous outbreaks [2], [4]. These climate–disease interactions are typically non-linear and time-lagged, complicating efforts based on classical linear models.

Traditional analytical approaches for dengue prediction in Bangladesh and South Asia have relied on regression-based or compartmental epidemiological models such as SIR and SEIR, sometimes combined with ARIMA-type time-series methods [5]. Although these models provide theoretical insights, assumptions of linearity, homogeneous mixing and fixed parameters limit their ability to represent delayed climatic effects and spatial heterogeneity in long, high-resolution surveillance datasets. As a result, prediction performance deteriorates when such models are applied to national-scale, division-wise daily data.

Machine learning (ML) offers flexible, data-driven alternatives capable of learning intricate non-linear relationships from multivariate time series [6]. Ensemble-based methods, particularly gradient boosting and random forests, have demonstrated strong predictive performance for infectious diseases including dengue [7]. However, ML models are often criticized as black boxes, raising concerns about transparency and trust for high-stakes public health decisions. Explainable AI (XAI) methods such as SHAP can bridge this gap by quantifying feature contributions and aligning model behaviour with epidemiological knowledge [8].

Existing ML-based dengue forecasting studies in Bangladesh have focused on specific cities, short time horizons or monthly data and often use random train–test splits that ignore temporal causality [9], [10]. Few works integrate division-wise daily incidence with high-resolution climate data in a unified spatio-temporal framework or systematically address interpretability. This motivates the development of a robust explainable ensemble framework adapted specifically to Bangladesh.

This study develops such a framework using daily DGHS surveillance and meteorological data from 2022 to 2025. The key contributions are:

- Construction of an integrated daily dengue–meteorological dataset covering all eight divisions with rigorous quality control and temporal alignment using DGHS, Open-Meteo and ERA5 sources [11]–[13]
- Design of a structured, time-aware ML pipeline with advanced temporal and climatic feature engineering and strict chronological validation [3]
- Systematic benchmarking of six regression models, identifying LightGBM as the most accurate and robust ensemble learner for national-scale dengue forecasting [14], [15]

- Application of SHAP-based XAI to interpret model behaviour and quantify the relative importance of lagged incidence, seasonal structure and climatic drivers in dengue outbreaks [8]

II. Methodology

A. Workflow

Fig. 1 summarizes the end-to-end workflow of the proposed dengue forecasting system, including data collection, integration, preprocessing, model training, chronological validation and explainability analysis. Daily division-wise dengue case counts are obtained from DGHS, while meteorological variables are retrieved from Open-Meteo and ERA5 using script-based queries [12], [13]. Necessary preprocessing steps—including data cleaning, temporal alignment, feature engineering and leakage control—are then applied. The integrated dataset is partitioned using a strictly chronological hold-out strategy, with January 2022–December 2024 used for training and January 2025–October 2025 used for testing. Baseline machine learning models are trained on the training set, hyperparameters are optimized using time-aware cross-validation and the best-performing model is selected based on MAE, RMSE, RMSLE and R^2 metrics. Finally, the selected model is evaluated on the unseen test set and interpreted using SHAP-based explainable AI methods.

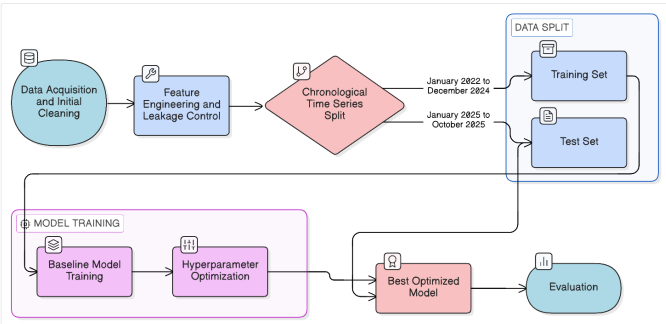

Fig. 1. Working procedure of the proposed dengue case prediction system.

B. Dataset

Daily dengue case counts were obtained from the DGHS surveillance database for each of the eight administrative divisions from 15 January 2022 to 31 October 2025 [11]. These official division-wise statistics constitute the primary target variable for model building and evaluation.

Meteorological data were collected using Python scripts that queried the Open-Meteo archive API and ERA5 reanalysis products [12], [13]. For each division, centroid latitude and longitude coordinates were used to extract daily maximum, minimum and mean temperature, rainfall, relative humidity and derived climate indices. All data processing was performed programmatically to ensure reproducibility and consistent coverage across divisions.

After integration and temporal alignment, the dataset forms a multivariate spatio-temporal time series with 11 088 daily observations across the eight divisions, including epidemiological, climatic, seasonal and engineered temporal features.

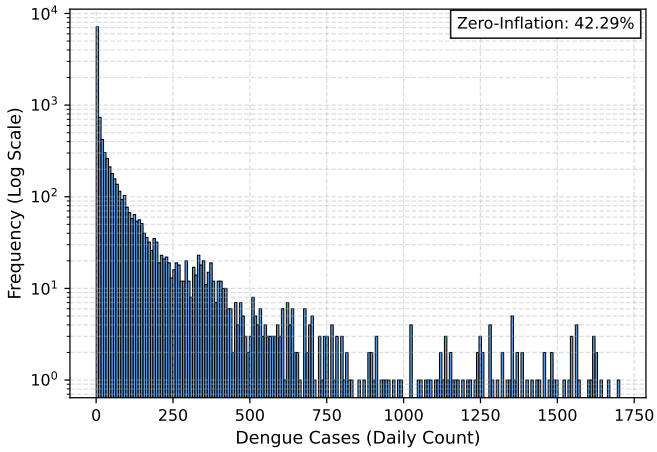

Fig. 2. Distribution of daily dengue cases in Bangladesh (2022–2025) in raw counts and log-transformed scale, highlighting heavy right skew and zero inflation.

C. Dataset Preprocessing

Exploratory data analysis revealed strong seasonality and inter-annual variability, with peaks during monsoon and post-monsoon and extended low-incidence periods in winter [2], [4]. The distribution of daily dengue cases is highly right-skewed and zero-inflated, with many zero-case days and extreme outbreaks exceeding 1 500 cases per day. Fig. 2 shows the distribution in raw and log-transformed scales.

A short gap in 2024 occurs due to a nationwide network outage that interrupted official surveillance and meteorological broadcasting. For this period, missing values are imputed using division-wise monthly averages to preserve regional differences without introducing look-ahead bias, while shorter gaps are filled using forward fill that respects chronological order. Continuous covariates required by certain models (e.g., SVR) are standardized based on training-data statistics only.

To handle skewness and zero inflation, the target variable is transformed using

$$y'_t = \log(y_t + 1), \tag{1}$$

where y_t denotes the original daily dengue count at time t. During evaluation, predictions are back-transformed via

$$\hat{y}_t = \exp(\hat{y}'_t) - 1 \tag{2}$$

to recover daily case counts.

D. Feature Engineering

Feature engineering encodes temporal dependence, seasonality, climate–disease interactions and spatial heterogeneity.

1) Autoregressive Lags: Autoregressive lag features capture dependence on past incidence:

$$y_t^{(\text{lag}n)} = y_{t-n}, \quad n \in \{3, 7, 14\}, \tag{3}$$

representing dengue counts three, seven and fourteen days prior to time t.

2) Rolling Window Statistics: Rolling-window statistics smooth short-term variability and highlight outbreak momentum. The 7-day and 14-day rolling means are defined as

$$\text{mean}_t^{(w)} = \frac{1}{w} \sum_{i=t-w+1}^{t} y_i, \quad w \in \{7, 14\}, \tag{4}$$

where y_i denotes the log-transformed case count at day i.

3) Seasonal Encoding: Seasonal patterns are encoded using both categorical seasons (Winter, Spring, Pre-Monsoon, Monsoon, Post-Monsoon) and Fourier-based harmonic terms. The seasonal signal is represented as

$$S_t = \sum_{k=1}^{K} \left[A_k \cos\left(\frac{2\pi kt}{T}\right) + B_k \sin\left(\frac{2\pi kt}{T}\right) \right], \tag{5}$$

where T is the annual period and K the number of harmonics.

4) Climate Indices: To capture non-linear climate effects, composite indices are constructed from temperature and humidity or rainfall. A simplified Heat Index (HI) can be expressed as

$$\text{HI} = c_1 + c_2 T + c_3 H + c_4 TH + c_5 T^2 + c_6 H^2 + c_7 T^2 H + c_8 T H^2, \tag{6}$$

where T is temperature, H is relative humidity and c_i are constants. A Rain–Temperature Index (RTI) combines rainfall and temperature:

$$\text{RTI} = \text{Rainfall} \times T. \tag{7}$$

5) Spatial Features: Spatial heterogeneity is represented via one-hot encoding of division identifiers, enabling a single global model to learn region-specific effects across all divisions.

E. Experimental Setup and Models

To emulate real-world forecasting, the dataset is partitioned chronologically: January 2022–December 2024 data are used for training, while January 2025–October 2025 data are reserved for testing. Time-series cross-validation within the training period is used for hyperparameter tuning, ensuring that each validation fold only uses future months relative to its training subset. All lagged and rolling features are constructed using only past information up to each prediction date.

Six regression models are evaluated:

- Ridge Regression,
- Random Forest,
- Support Vector Regression (SVR),
- XGBoost [15],
- CatBoost,
- LightGBM [14].

Hyperparameters are optimized using randomized search with five-fold time-series cross-validation on the training period, targeting RMSLE in the log domain and employing early stopping for boosting models. After tuning, each model is refitted on the full training set and evaluated on the held-out test period.

TABLE I
BASELINE MODEL PERFORMANCE WITH DEFAULT HYPERPARAMETERS (TEST PERIOD 2025)

Model	R^2	MAE	RMSE	RMSLE
LightGBM	**0.899**	**9.148**	**20.236**	**0.773**
CatBoost	0.872	9.852	22.739	0.777
XGBoost	0.849	10.759	24.708	0.812
Random Forest	0.896	9.173	20.476	0.778
Ridge	0.640	17.206	38.203	1.097
SVR	0.689	14.389	35.478	0.957

TABLE II
PERFORMANCE METRICS OF THE APPLIED MODELS WITH OPTIMIZED HYPERPARAMETERS (TEST PERIOD 2025)

Model	R^2	MAE	RMSE	RMSLE
LightGBM	**0.900**	**8.927**	**20.094**	**0.765**
CatBoost	0.898	9.026	20.310	0.768
XGBoost	0.891	9.366	20.998	0.772
Random Forest	0.888	8.984	21.234	0.769
Ridge	0.637	17.275	38.322	1.097
SVR	0.383	18.702	49.997	0.910

Model performance is assessed via MAE, RMSE, RMSLE and R^2, providing complementary views of absolute error, error sensitivity to large outbreaks, relative accuracy and explained variance. SHAP values are computed for the final LightGBM model to provide global and local explanations of predictions.

III. RESULT ANALYSIS

Experiments are implemented in Python using scikit-learn and gradient-boosting libraries on a local workstation. Baseline model performance using default hyperparameters and the impact of tuning are summarised in Tables I and II.

Hyperparameter tuning substantially improves ensemble models, especially LightGBM and CatBoost, while Ridge and SVR show limited or negative gains, indicating that they are ill-suited to the highly non-linear, high-dimensional structure of the data. LightGBM emerges as the most accurate and robust model, achieving the highest R^2 and the lowest MAE, RMSE and RMSLE among all candidates.

A. Temporal Fit and Error Trends

Fig. 3 presents a time-aligned comparison of observed and LightGBM-predicted daily dengue cases for the 2025 test period, along with the absolute error, to visualize tracking performance during peaks and troughs. Predicted trajectories closely track outbreak dynamics, capturing seasonal peaks and troughs, with the largest errors during extreme monsoon and post-monsoon surges where incidence fluctuates rapidly.

B. Spatio-Temporal Variability

The division-wise predicted case counts in Fig. 4 highlight both spatial heterogeneity and seasonal variability in dengue dynamics. Dhaka and Barishal exhibit the highest predicted burdens, particularly during Monsoon and Post-Monsoon,

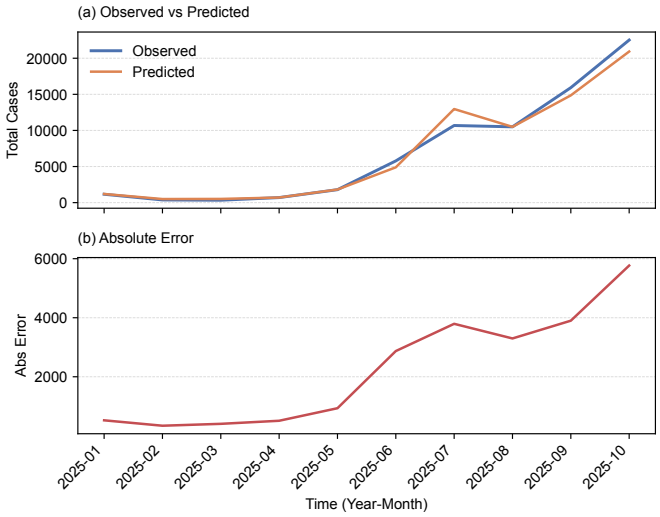

Fig. 3. Observed vs. LightGBM-predicted daily dengue cases and absolute error during the 2025 test period.

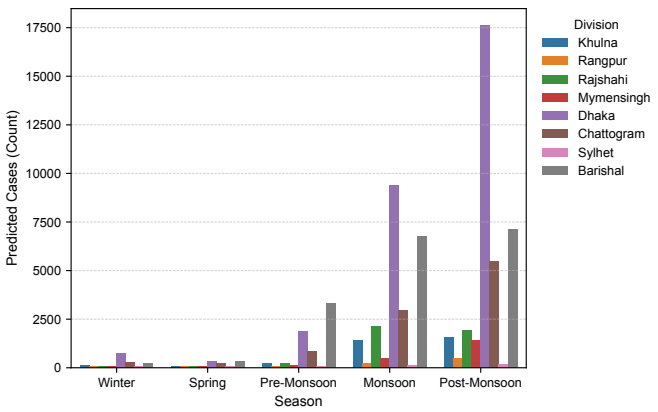

Fig. 4. Spatio-temporal prediction of dengue cases by division and season for 2025 using LightGBM.

whereas Mymensingh and Rangpur remain low and stable, reflecting underlying differences in urbanization, climate and population density.

C. Seasonal and Regional Error Variability

Seasonal Mean Absolute Error (MAE) by division is summarized in Fig. 5. Errors are highest in Dhaka during Monsoon and Post-Monsoon, reflecting complex urban transmission dynamics and extreme outbreak intensity, and lowest in Mymensingh and Rangpur, where incidence is low and stable.

IV. Explainable AI and Discussion

A. Climatic Drivers of Dengue

Stacked climate trend plots in Fig. 6 relate dengue case dynamics to temperature and rainfall for 2025. Incidence peaks align with periods of sustained high temperatures and heavy monsoon rainfall, consistent with the role of warm, wet conditions in boosting mosquito populations and viral replication.

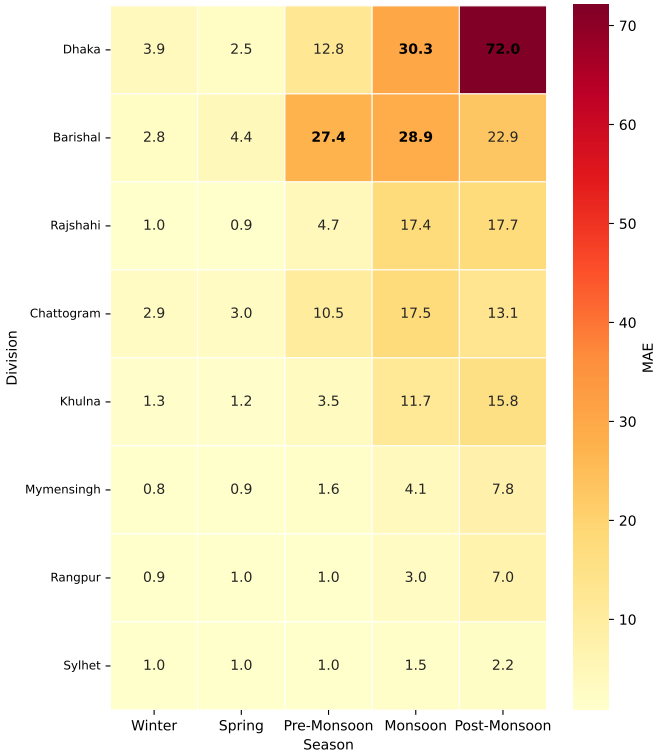

Fig. 5. Division-wise seasonal MAE of LightGBM, showing higher errors during Monsoon and Post-Monsoon in high-burden divisions.

B. Feature Importance via SHAP

SHAP values are computed for the optimized LightGBM model to quantify the contribution of individual predictors. The beeswarm plot in Fig. 7 shows that lagged dengue incidence (3, 7 and 14 days) and rolling averages are the strongest predictors, confirming the autoregressive nature of dengue transmission. Seasonal Fourier encodings and climatic variables (temperature and rainfall indices) also show meaningful contributions, supporting their role as environmental modulators of risk.

These XAI results demonstrate that the model bases its forecasts on epidemiologically plausible drivers and provide actionable insights into which features most influence predicted risk across regions and seasons.

V. Conclusion

This study develops an explainable ensemble-learning framework for spatio-temporal dengue outbreak forecasting across Bangladesh's eight administrative divisions by integrating daily DGHS surveillance data (Jan 2022–Oct 2025) with high-resolution Open-Meteo and ERA5 meteorological datasets. Through leakage-aware preprocessing, advanced feature engineering (autoregressive lags, rolling statistics, Fourier seasonal terms, and climate indices), and strict chronological validation, LightGBM achieved superior predictive performance (R^2 = 0.900, MAE = 8.927, RMSE = 20.094,

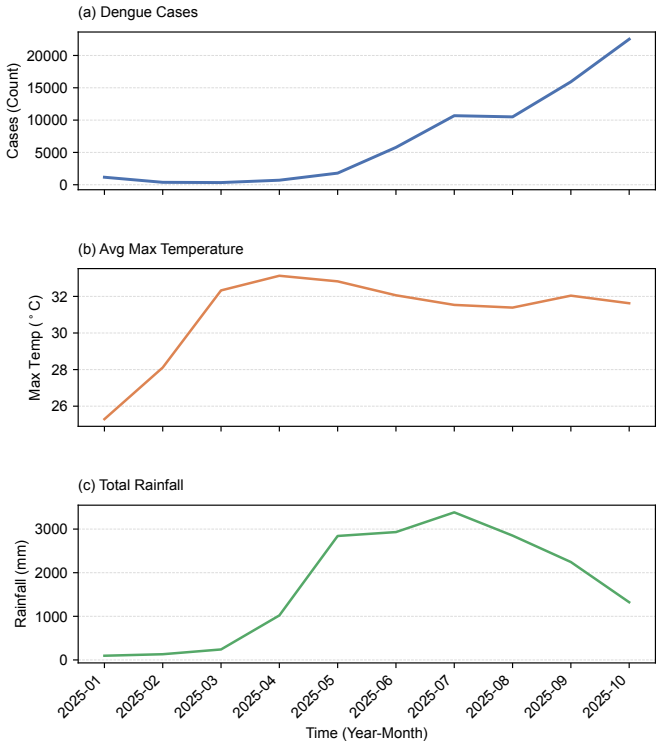

Fig. 6. Stacked climate trends (daily dengue cases, mean temperature and rainfall) for 2025, illustrating alignment between incidence peaks and warm, wet conditions.

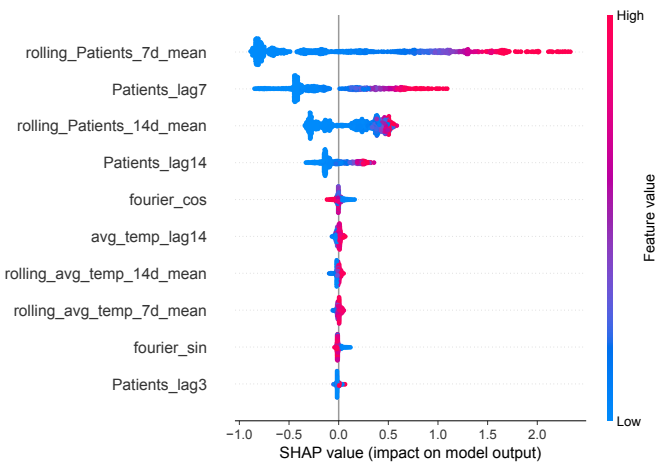

Fig. 7. SHAP beeswarm plot of global feature importance for LightGBM model. Lagged incidence and rolling averages dominate; seasonal/climatic factors contribute secondarily.

RMSLE = 0.765) on the 2025 test period, outperforming other ensemble baselines.

SHAP-based interpretability analysis confirms epidemiological plausibility, demonstrating that recent incidence lags and short-term rolling averages dominate model predictions, followed by temperature and rainfall indices as well as seasonal harmonics. The proposed framework supports climate-informed early warning systems, targeted vector control in high-burden divisions such as Dhaka and Barishal, and optimized healthcare resource allocation during monsoon and post-monsoon peaks.

Future work will focus on finer spatial granularity (upazila-level forecasting), integration of human mobility patterns, incorporation of real-time surveillance streams, and inclusion of climate change projections to enable long-horizon forecasting under warming scenarios.

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